# Chapter 7. Matching and Instrumental Variables

Nearest-neighbour matching, Abadie–Imbens inference, IV estimators, compliers and the LATE

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## 7.1 Matching

**Motivation.** The central question is how to estimate the effect of treatment on the treated by imputing each treated unit's missing control outcome from similar control units, and how to design such comparisons so that the result does not hinge on modeling choices.

### 7.1.1 Matching as nonparametric imputation

**Motivation.** Motivate matching as a model-free way to impute counterfactual control outcomes.

#### 7.1.1.1 Setup

$N$ units, treatment $X_i \in \{0,1\}$, covariates $Z_i$, potential outcomes $Y_i(0), Y_i(1)$; $N_1 = |\{i : X_i = 1\}|$ treated, $N_0 = N - N_1$ controls. Assumptions: ignorability $(Y(0), Y(1)) \perp\!\!\!\perp X \mid Z$ and positivity $0 < P(X = 1 \mid Z) < 1$. Focus: the ATT.

#### 7.1.1.2 From regression to matching

Under ignorability, with $\mu_t(Z) = \mathbb{E}[Y \mid X = t, Z]$,

$$\text{ATT} = \mathbb{E}_{Z|X=1}\big[\underbrace{\mu_1(Z)}_{\text{observed}} - \underbrace{\mu_0(Z)}_{\text{impute}}\big].$$

Regression performs model-based imputation of $\mu_0(Z_i)$:

$$\hat{\tau}_{\text{reg}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i(Y_i - \hat{\mu}_0(Z_i)).$$

When the groups differ in $Z$, the estimate depends heavily on the functional form of $\hat{\mu}_0$ (model dependence, Section 7.1.5). Matching performs *nonparametric* imputation instead.

#### 7.1.1.3 The matching idea

Often there is a large reservoir of potential controls, so each treated unit can be matched to one or more distinct controls with similar covariates.

**Matching as study design.** Matching is a design tool, not only an analysis technique. In many observational studies outcomes are not yet measured when matching is done: identify treated units, select which controls to recruit/follow, then measure outcomes for both. This separates the design stage (whom to compare) from the analysis stage (how to estimate); design choices are made without seeing outcomes, removing the temptation to tune the analysis toward a desired result (Stuart 2010, §1).

**Bias–variance trade-off in $M$.** Each treated unit may be matched to $M$ controls. $M = 1$: lower bias, higher variance. $M = 3$: farther matches increase bias, while averaging three outcomes reduces variance (Abadie and Imbens 2006).

#### 7.1.1.4 The matching estimator

Let $\mathcal{M}_i$ be the matched set of treated unit $i$ (units with covariates similar to $i$'s). Still assuming ignorability,

$$\hat{\tau}_{\text{match}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i\Big(Y_i - \underbrace{\frac{1}{|\mathcal{M}_i|}\sum_{j\in\mathcal{M}_i} Y_j}_{\approx Y_i(0)}\Big).$$

Matching can be one-to-one or one-to-many, with or without replacement, etc.

#### 7.1.1.5 Exact matching and coarsened exact matching

**Exact matching:** for each treated $i$ find a control $j$ with $Z_j = Z_i$. It ensures perfect balance and no model dependence but is usually infeasible with many or continuous covariates.
**Coarsened exact matching** (CEM; Iacus et al. 2012): discretize covariates (categories may have substantive meaning), then match exactly on the coarsened values. Some treated units may still have no matches (lack of overlap).

### 7.1.2 Nearest-neighbor matching and its theory

**Motivation.** Define NN matching, compare matching algorithms, and state its large-sample behavior.

#### 7.1.2.1 Definition

Let $\mathcal{M}_i$ be the indices of the $M$ closest units of the opposite group in a distance $\|\cdot\|$. The imputed potential outcomes are

$$\hat{Y}_i(t) = \begin{cases} Y_i, & X_i = t, \\ \dfrac{1}{M}\sum_{j\in\mathcal{M}_i} Y_j, & X_i = 1-t, \end{cases} \qquad t \in \{0,1\},$$

and the ATT estimator is $\hat{\tau}_{\text{ATT}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i(Y_i - \hat{Y}_i(0))$.

#### 7.1.2.2 Algorithms (matching without replacement)

- **Greedy:** order the treated units (e.g. hardest-to-match first); give each the $M$ closest remaining controls, set them aside, repeat.
- **Optimal:** build the $N_1 \times N_0$ distance matrix $D$; choose entries so that each row has exactly $M$ entries and each column at most one, minimizing the total distance. The Hungarian algorithm solves this in polynomial time.

**Why order matters.** With $M = 1$ and the same data, an arbitrary order lets $i_2$ take the only control near the isolated $i_3$ (distance 0.9), forcing $i_3$ to a control 2.9 away (worst distance 2.9). Processing the hardest unit first, $i_3$ secures its nearby control (1.2), $i_2$ takes another at 1.1, and the worst distance drops to 1.2.

#### 7.1.2.3 The curse of dimensionality

In one dimension $N$ uniform points have spacing about $1/N$; in $p$ dimensions about $N^{-1/p}$ per axis, so the nearest-neighbor distance is $O(N^{-1/p})$. If $\mu_0$ is Lipschitz with constant $L$,

$$|\mu_0(Z_i) - \mu_0(Z_j)| \le L \, \|Z_i - Z_j\| = O(N^{-1/p}),$$

which is the bias from mismatch. The rate slows dramatically with $p$:

| $p$ | 1 | 5 | 10 | 20 |
|---|---|---|---|---|
| $N^{-1/p}$ for $N = 10{,}000$ | 0.0001 | 0.16 | 0.40 | 0.63 |

This motivates dimension reduction, e.g. propensity-score matching (Abadie and Imbens 2006, Thm. 1).

#### 7.1.2.4 ATT matching: independence and consistency

For NN matching with fixed $M$ and replacement, the ATT imputes $\hat{Y}_i(0)$ for each treated unit from controls; each treated unit appears exactly once as a subject, so the imputed counterfactuals are independent across treated units. Independence plus the law of large numbers (and the bias being negligible when the control reservoir is large) give $\sqrt{N}$-consistency for the ATT. Matching is most natural for ATT estimation (Abadie and Imbens 2006, Thms. 1–4).

#### 7.1.2.5 A toy example

$N_1 = 3$ treated, $N_0 = 6$ controls, $p = 2$, $M = 1$, distance $\|\cdot\|_1$:

| | Treated | | | | Control | | |
|---|---|---|---|---|---|---|---|
| Unit | $Z_1$ | $Z_2$ | $Y$ | Unit | $Z_1$ | $Z_2$ | $Y$ |
| $i = 1$ | 3 | 5 | 8 | $j = 1$ | 1 | 2 | 2 |
| $i = 2$ | 6 | 2 | 12 | $j = 2$ | 2 | 6 | 4 |
| $i = 3$ | 7 | 7 | 10 | $j = 3$ | 4 | 1 | 6 |
| | | | | $j = 4$ | 3 | 4 | 5 |
| | | | | $j = 5$ | 5 | 3 | 7 |
| | | | | $j = 6$ | 8 | 6 | 9 |

Matches: $i = 1 \to j = 4$ (distance 1), $i = 2 \to j = 5$ (2), $i = 3 \to j = 6$ (2). Hence

$$\hat{\tau}_{\text{ATT}} = \tfrac{1}{3}\big[(8 - 5) + (12 - 7) + (10 - 9)\big] = 3.0.$$

The matches are imperfect ($Z_2 - Z_5 \neq 0$), and the mismatch creates bias, which the next estimator corrects.

#### 7.1.2.6 Bias-corrected matching (Abadie and Imbens 2011)

NN matching uses $Y_j$ to estimate $Y_i(0)$, but $Y_j$ approximates $\mu_0(Z_j)$, not $\mu_0(Z_i)$. Adjust each control outcome by the predicted difference due to mismatch:

$$\hat{\tau}_{\text{bc}} = \frac{1}{N_1} \sum_{i:X_i=1} \Big( Y_i - \frac{1}{M} \sum_{j \in \mathcal{M}_i} \big[ Y_j + \hat{\mu}_0(Z_i) - \hat{\mu}_0(Z_j) \big] \Big),$$

where $\hat{\mu}_0$ is a regression fit of $\mathbb{E}[Y \mid Z, X = 0]$. Naive matching uses $Y_j$ as $Y_i(0)$, ignoring the slope of $\mu_0$ between $Z_j$ and $Z_i$; the correction shifts by $\hat{\mu}_0(Z_i) - \hat{\mu}_0(Z_j)$. This is the default in the R packages `MatchIt` and `Matching`.

#### 7.1.2.7 Inference after matching

Variance estimator for the ATT (Abadie and Imbens 2006):

$$\widehat{\mathrm{Var}}(\hat{\tau}_{\mathrm{ATT}}) = \frac{1}{N_1^2}\sum_{i=1}^{N}\big(1 + K_M(i)\big)^2\hat{\sigma}_i^2,$$

where $K_M(i)$ is the number of times unit $i$ is used as a match and $\hat{\sigma}_i^2$ estimates $\mathrm{Var}(Y_i \mid Z_i, X_i)$. Units matched often inflate the variance, analogous to extreme IPW weights. **The bootstrap is invalid** for NN matching: bootstrap validity requires small data changes to cause small estimate changes, but the identity of the matched unit is a discontinuous function of the data (Abadie and Imbens 2008).

#### 7.1.2.8 Kernel matching

All controls within a bandwidth $h$ get positive weight, closer ones more. With a kernel $K$ (e.g. Gaussian $K(u) = e^{-u^2/2}$),

$$w_i(j) = \frac{K\Big(\frac{Z_j - Z_i}{h}\Big)}{\sum_{k:X_k=0} K\Big(\frac{Z_k - Z_i}{h}\Big)}, \qquad \hat{\tau}_{\mathrm{kernel}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i\Big(Y_i - \sum_{j:X_j=0} w_i(j)Y_j\Big).$$

For large $p$ kernel matching faces the same curse of dimensionality.

### 7.1.3 Propensity-score matching

**Motivation.** Reduce the matching problem to one dimension and understand the caveats.

**Dimension reduction.** NN bias grows like $O(N^{-1/p})$; with 20 binary covariates there are $2^{20} \approx 10^6$ patterns and direct matching is nearly impossible. The propensity score $e(Z) = P(X = 1 \mid Z)$ reduces $p$ dimensions to one, and $Y(0), Y(1) \perp\!\!\!\perp X \mid Z$ implies $Y(0), Y(1) \perp\!\!\!\perp X \mid e(Z)$.

**Distance.** $d(Z_i, Z_j) = |\hat{e}(Z_i) - \hat{e}(Z_j)|$, or (preferred) on the logit scale

$$d(Z_i, Z_j) = |\mathrm{logit}\,\hat{e}(Z_i) - \mathrm{logit}\,\hat{e}(Z_j)|, \qquad \mathrm{logit}(p) = \ln\frac{p}{1-p}.$$

Distances that are small on the probability scale are compressed near 0 and 1; the logit stretches them out.

**Specification: balance, not fit.** The PS is usually estimated by logistic regression, but GBMs, random forests, etc. are also used. The goal is covariate balance, not accurate prediction of treatment. Collinearity is not a concern (coefficients are not interpreted); overfitting to the sample can even help (Appendix). Specify iteratively: check balance on covariates, squares and interactions; if imbalanced, add these terms and re-estimate. With missing covariate values, impute and include missingness indicators (Stuart 2010; Rosenbaum and Rubin 1984).

**Common support.** Each treated unit needs controls with similar covariates. Units outside the overlap region of the two PS distributions have no good matches and should be trimmed.

**Caveats.** Matching on the *estimated* PS can be more efficient than on the true PS (Abadie and Imbens 2016): estimation noise adapts to the sample, so matched units are closer in covariate space; this relies on correct specification. King and Nielsen (2019) describe the "PS paradox": a narrower PS caliper improves PS balance but can worsen covariate balance, because units with similar $\hat{e}(Z)$ can have very different $Z$. Always check multivariate covariate balance, not just PS balance.

### 7.1.4 Practical matching

**Motivation.** Describe the iterative matching workflow and its design choices.

**Workflow.** Choose a distance → perform matching → check balance → if imbalanced, adjust (metric, $M$, caliper) and repeat; if balanced, estimate and do inference. After matching, regression or weighting on the matched sample can correct residual imbalance.

**Design choices.**

- Which covariates to include (recall the adjustment criterion; never include post-treatment variables).
- Distance metric: Mahalanobis, propensity score, or hybrid.
- Number of matches $M$: fixed or varying (caliper-based). Smaller $M$: smaller bias, larger variance; larger $M$: the reverse. With a much larger control group, 1-to-many matching is natural.
- With or without replacement.

**Mahalanobis distance** accounts for correlations and scales: $d(x, y) = \sqrt{(x-y)^\top S^{-1}(x-y)}$ with $S$ the sample covariance (treated group or pooled). With few continuous covariates ($p \lesssim 5$–$8$) it works well; with many covariates or a complex treatment model the PS gives effective dimension reduction. Common practice: match on the PS with a caliper, then refine with Mahalanobis distance within the caliper.

**Caliper matching.** With caliper width $c$, all controls within distance $< c$ of a treated unit are potential matches, so $M$ varies by unit (e.g. $M_1 = 2$, $M_2 = 3$, $M_3 = 1$). A common choice is $c = 0.2 \times \mathrm{SD}(\mathrm{logit}\,\hat{e}(Z))$ (Austin 2011, simulation-based).

**Full matching** (Rosenbaum 1991; Hansen 2004). Standard matching discards unmatched controls. Full matching places every unit into a matched set with variable ratio (some 1:$k$, others $k$:1), minimizing total within-set distance (solved efficiently by software). It uses more controls within the overlap and produces a weighted sample, a bridge between matching and IPW. Often combined with a caliper. In R: `MatchIt` with `method = "full"`.

**With versus without replacement.** With replacement: computationally easier, not order-dependent, but some units (especially with extreme PS) may be matched many times and dominate (like extreme IPW weights). Without replacement: avoids extreme reuse but is order-dependent (greedy vs. optimal matters). Software: `MatchIt` (Ho et al.), `Matching` (Sekhon).

**Checking balance.** Standardized mean difference of covariate $k$:

$$\mathrm{SMD}_k = \frac{1}{s_t} \cdot \frac{1}{N_1} \sum_{i=1}^{N} X_i \Big( Z_{ik} - \frac{1}{|\mathcal{M}_i|} \sum_{j \in \mathcal{M}_i} Z_{jk} \Big),$$

with $s_t$ the SD of covariate $k$ in the treated group before matching. Compare whole distributions via empirical CDFs and Kolmogorov–Smirnov statistics. **Do not use $p$-values** for balance: they conflate changes in balance with changes in sample size, and balance is an in-sample property (Ho et al. 2007).

**Love plot.** A dot plot of SMDs per covariate before and after adjustment, with reference lines at $\pm 0.1$. In the LaLonde (1986) job-training data (propensity score, age, education, married, no degree, race indicators, earnings in 1974 and 1975), unadjusted SMDs range up to about 1.5–2 in absolute value; after matching they fall inside $\pm 0.1$.

### 7.1.5 Model dependence and empirical evidence

**Motivation.** Explain why matching reduces sensitivity to modeling assumptions, and what the LaLonde benchmark teaches.

**The extrapolation problem.** When treated and control groups differ on covariates, regression must extrapolate into sparse regions. A linear and a quadratic specification can fit the control region equally well yet give very different values in the treated region. Model dependence is greatest where data are sparse (Ho, Imai, King and Stuart 2007).

**Matching as preprocessing.** After matching, the distributions overlap and regression interpolates rather than extrapolates; different specifications give similar estimates. Workflow:

(1) match for balance, (2) regress/weight on the matched sample, (3) obtain estimates robust to model choice. "Matching is the starting point, not the end."

**The LaLonde benchmark.** The National Supported Work (NSW) program was a randomized job-training experiment (LaLonde 1986), with experimental ATT $\approx$ \$1,794. Replacing the experimental controls with observational ones (CPS/PSID), unadjusted estimates range from $-$\$15,000 to $+$\$2,000, because the observational controls differ dramatically in demographics, prior earnings and employment. It is the canonical testbed for matching methods.

**Lessons.** Dehejia and Wahba (1999): PS matching on a subsample with pre-treatment earnings recovers the experimental benchmark. Smith and Todd (2005): the result is fragile; different subsamples, matching methods and distance metrics lead to different conclusions. A deeper issue: the DW subsample included pre-treatment earnings, arguably the most important confounder; without it, ignorability itself is doubtful whatever the method. Key lessons: covariate sufficiency comes before method choice; matching can recover experimental benchmarks, but with no guarantees.

**Practical takeaways.**

- Matching is a nonparametric preprocessing step, not an end in itself; then apply regression or weighting on the matched sample.
- Always check multivariate covariate balance, not just PS balance.
- In high-dimensional settings combine matching with regression.
- Sensitivity analysis for unmeasured confounding is an important complement.

### 7.1.6 Appendix: ATE matching: why it is harder

For the ATE one must impute $\hat{Y}_i(0)$ for treated and $\hat{Y}_i(1)$ for control units. Each unit serves both as a subject and as a match for others, creating cross-group dependence absent for the ATT or ATC alone. Moreover the bias $O(N^{-1/p})$ dominates the $O(N^{-1/2})$ sampling error when $p > 2$: $\text{Bias}(\hat{\tau}_{\text{ATE}}) = O(N^{-1/p}) \gg O(N^{-1/2})$, so simple matching is not $\sqrt{N}$-consistent for the ATE. Use bias correction (Abadie and Imbens 2011) or IPW/DR instead (Abadie and Imbens 2006).

### 7.1.7 Appendix: Why the estimated PS can beat the true PS

The true $e(Z)$ is a population quantity; two units can have $e(Z_i) = e(Z_j)$ yet very different covariates. The estimated $\hat{e}(Z) = g(Z^\top \hat{\beta})$ is fit to this sample, so $\hat{e}(Z_i) \approx \hat{e}(Z_j)$ means $Z_i, Z_j$ are close in the directions that distinguish treated from control in this sample: estimation noise acts as implicit covariate refinement. Under correct specification (Abadie and Imbens 2016), $\text{Var}(\hat{\tau}_{\text{ATT}} \mid \hat{e}) \leq \text{Var}(\hat{\tau}_{\text{ATT}} \mid e)$. Analogously, IPW with $\hat{e}$ attains the semiparametric efficiency bound while IPW with $e$ does not (Hirano, Imbens and Ridder 2003). The advantage disappears under misspecification.

### 7.1.8 Appendix: The PS paradox (King and Nielsen 2019)

Many covariate configurations map to the same PS. Unit A (treated) with high income and low education and unit B (control) with low income and high education may both have $\hat{e} = 0.50$: perfect PS balance, opposite covariates. A tighter PS caliper yields more "PS-similar but covariate-different" pairs, so a better PS match can mean worse covariate balance; the remaining imbalance may require more extrapolation than the raw data, increasing model dependence.

### 7.1.9 Appendix: PS overfitting and the broader picture

Overfitting the PS model can help: the goal is in-sample balance, and a flexible model captures sample-specific patterns so that similar $\hat{e}$ means closer $Z$. The limit is overlap: if $\hat{e} \to 0$ or 1 for all units, overlap is destroyed. Three results, one message: the estimated PS beats the true PS (sample adaptation), overfitting helps (the same mechanism), but the PS paradox shows that 1D compression can still lose information. The PS is a dimension-reduction tool, not a substitute for checking balance. Practical resolution: use the PS for dimension reduction (especially for large $p$), always verify multivariate balance, and consider Mahalanobis matching within a PS caliper.

### 7.1.10 Appendix: Beyond binary treatment

With multiple treatment levels (doses, programs): compare pairs by standard binary matching, or use the generalized propensity score (GPS) for $K$ levels (Imai and van Dyk 2004). For continuous treatments the GPS is the conditional density of treatment given covariates (Hirano and Imbens 2004). In practice subclassification or weighting on the GPS is more common than matching.

### 7.1.11 Appendix: Bias-corrected matching versus post-matching regression

In bias-corrected matching, regression is a correction term inside each matched pair; matching is primary and regression fixes the residual $Z_i \neq Z_j$ gap. In post-matching regression, matching is preprocessing (balanced sample) and a regression $Y \sim X + Z$ on the matched data does the estimation. Why not regress on the full sample? $\hat{\mu}_0$ can be learned from all controls, and balance is irrelevant if the model is correct; but if the model is wrong, full-sample regression extrapolates and is model-dependent.

| | Model correct | Model wrong |
|---|---|---|
| Full-sample regression | optimal (more data) | risky (extrapolation) |
| Post-matching regression | less efficient (data discarded) | safe (interpolation) |

If controls lie in $Z \in [0, 3]$ and treated in $[5, 8]$, full regression must extrapolate $\hat{\mu}_0(6)$: a linear model gives answer A, a quadratic answer B. After matching, controls also lie near $[5, 8]$ and both models give similar answers. This is the core of "matching as preprocessing" (Ho et al. 2007).

### 7.1.12 Appendix: Stratification

With a single discrete covariate $Z$ with levels $k$ that makes assignment ignorable, $\mathbb{E}[Y(1)] = \sum_k \mathbb{E}(Y \mid Z = k, X = 1)P(Z = k)$. With $n_k$ units in cell $k$ and cell means $\bar{Y}_{k,x}$, estimate $\mathbb{E}[Y(1)]$ by $\sum_k \bar{Y}_{k,1} n_k/n$, hence

$$\hat{\tau} = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) \frac{n_k}{n}.$$

For continuous $Z$, split it into $k$ classes: $\hat{\tau}_k = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) n_k/n$, generally biased for $\tau$. Let $R_k = 1 - \frac{\mathbb{E}(\hat{\tau}_k) - \tau}{\mathbb{E}(\hat{\tau}_1) - \tau}$ be the fraction of bias removed by $k$-stratum stratification. Cochran (1968):

$R_k \geq 90\%$ for $k \geq 5$ across a broad class of models, so stratifying into five or more blocks removes roughly 90% of the confounding bias.

## 7.2 Instrumental variables

**Motivation.** The central question is how a causal effect can be learned when confounders are unmeasured and no adjustment set exists, by exploiting a variable that moves the treatment but affects the outcome only through it.

### 7.2.1 Motivation

**Motivation.** See instruments in action and state what makes an instrument good.

#### 7.2.1.1 Instrumental variables in action

| Study | Instrument $Z$ | Treatment $X$ | Outcome $Y$ |
|---|---|---|---|
| Angrist and Krueger (1991) | quarter of birth | years of schooling | earnings |
| Card (1995) | proximity to college | years of schooling | earnings |
| Acemoglu et al. (2001) | settler mortality | institutional quality | GDP per capita |
| Miguel et al. (2004) | rainfall shocks | economic growth | civil conflict |
| Levitt (1996) | overcrowding litigation | prison population | crime rate |

**Returns to education.** Ability $U$ confounds schooling $X$ and earnings $Y$. Angrist and Krueger: compulsory-schooling laws interact with quarter of birth; those born early in the year can drop out sooner and get less schooling. Card: growing up near a four-year college lowers the cost of attending, so proximity leads to more education and, through it, higher earnings.
**Institutions and development.** Geography and culture $U$ confound institutions $X$ and GDP $Y$. Acemoglu, Johnson and Robinson: high settler mortality led to extractive institutions, low mortality to inclusive ones; the institutional differences persist and explain modern income gaps. Miguel et al. use rainfall as an instrument for economic growth to study civil conflict in Africa.

#### 7.2.1.2 Why instruments?

So far effects were identified by back-door adjustment, which requires measuring all confounders. If some are unmeasured, no adjustment set blocks the confounding. Idea: find $Z$ that affects $Y$ only through $X$ ($Z \to X \to Y$, $X \leftarrow U \to Y$, $Z \perp\!\!\!\perp U$). Such $Z$ is an *instrumental variable*.

#### 7.2.1.3 Conditions for a good instrument

IVs are very popular in economics, but good ones are hard to find. A good instrument must satisfy

1. **Relevance:** a strong effect on the treatment (large $\mathrm{Cov}(X, Z)$); otherwise the IV estimate has large variance;
2. **Exclusion restriction:** no direct effect on the outcome;

and (implicitly) no confounding between $Z$ and $Y$.

### 7.2.2 IV under linearity

**Motivation.** Derive the classical IV estimators in the linear constant-effect model.

#### 7.2.2.1 The econometric setup

Consider $Y = \beta_0 + \beta_1 X + \beta_2 W + \varepsilon$ with scalar treatment $X$ and covariates $W$; the estimand is $\beta_1$. $X$ is *endogenous* (correlated with $\varepsilon$, i.e. confounded) and $W$ is *exogenous* (uncorrelated with $\varepsilon$). Because $\varepsilon$ depends on $X$, OLS of $\beta_1$ is biased. An instrument is a $K$-vector $Z$ with $(Z, W) \perp \varepsilon$ (uncorrelatedness: $\mathrm{Cov}[\varepsilon, Z] = \mathrm{Cov}[\varepsilon, W] = 0$). If $K > 1$ the model is over-identified; if $K = 1$, just-identified.

#### 7.2.2.2 Just-identified, no covariates

**IV estimator.** The ratio of covariances:

$$\hat{\beta}_{1,\mathrm{iv}} = \frac{\widehat{\mathrm{Cov}}(Y_i, Z_i)}{\widehat{\mathrm{Cov}}(X_i, Z_i)} = \frac{\frac{1}{N}\sum_{i=1}^{N}(Y_i - \bar{Y})(Z_i - \bar{Z})}{\frac{1}{N}\sum_{i=1}^{N}(X_i - \bar{X})(Z_i - \bar{Z})}.$$

With a binary instrument this is the *Wald estimator*

$$\hat{\beta}_{1,\mathrm{iv}} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0},$$

where $\bar{Y}_z, \bar{X}_z$ are sample means within group $Z = z \in \{0, 1\}$ (since $\widehat{\mathrm{Cov}}(Y, Z) = \hat{p}(1-\hat{p})(\bar{Y}_1 - \bar{Y}_0)$ with $\hat{p}$ the fraction with $Z = 1$, and likewise for $X$).

**Indirect least squares (ILS).** Fit two reduced-form regressions

$$Y = \pi_{10} + \pi_{11} Z + \varepsilon_1, \qquad X = \pi_{20} + \pi_{21} Z + \varepsilon_2;$$

the ratio of OLS slopes gives $\hat{\beta}_{1,\mathrm{ils}} = \hat{\pi}_{11}/\hat{\pi}_{21} = \hat{\beta}_{1,\mathrm{iv}}$, since each slope is $\widehat{\mathrm{Cov}}(\cdot, Z)/\widehat{\mathrm{Var}}(Z)$.

**Two-stage least squares (2SLS).** Stage 1: regress $X$ on $Z$, $\hat{X} = \hat{\pi}_{20} + \hat{\pi}_{21} Z$. Stage 2: regress $Y = \beta_0 + \beta_1 \hat{X} + \varepsilon$ by OLS to get $\hat{\beta}_{1,\mathrm{2sls}}$. Intuition: the instrument recovers the "unconfounded portion" of $X$. Since $\widehat{\mathrm{Cov}}(Y, \hat{X}) = \hat{\pi}_{21}\widehat{\mathrm{Cov}}(Y, Z)$ and $\widehat{\mathrm{Var}}(\hat{X}) = \hat{\pi}_{21}^2 \widehat{\mathrm{Var}}(Z)$, $\hat{\beta}_{1,\mathrm{2sls}} = \widehat{\mathrm{Cov}}(Y, Z)/(\hat{\pi}_{21}\widehat{\mathrm{Var}}(Z)) = \hat{\beta}_{1,\mathrm{iv}}$. Hence $\hat{\beta}_{1,\mathrm{iv}} = \hat{\beta}_{1,\mathrm{ils}} = \hat{\beta}_{1,\mathrm{2sls}}$.

#### 7.2.2.3 Just-identified, with covariates

ILS: $Y = \pi_{10} + \pi_{11} Z + \pi_{12} W + \varepsilon_1$, $X = \pi_{20} + \pi_{21} Z + \pi_{22} W + \varepsilon_2$, $\hat{\beta}_{1,\mathrm{ils}} = \hat{\pi}_{11}/\hat{\pi}_{21}$. 2SLS: $\hat{X} = \hat{\pi}_{20} + \hat{\pi}_{21} Z + \hat{\pi}_{22} W$, then $Y = \beta_0 + \beta_1 \hat{X} + \beta_2 W + \varepsilon$; $\hat{\beta}_{1,\mathrm{2sls}}$ is the OLS coefficient of $\hat{X}$.

**IV derivation.** Given $X = \pi_{20} + \pi_{21} Z + \pi_{22} W + \varepsilon_2$ and $Y = \beta_0 + \beta_1 X + \underbrace{\beta_2 W + \varepsilon}_{\eta}$, assume in addition $\mathrm{Cov}[W, Z] = 0$. Then $\mathrm{Cov}[X, Z] \neq 0$ but $\mathrm{Cov}[\eta, Z] = 0$, so

$$\mathrm{Cov}[Y, Z] = \mathrm{Cov}[\beta_0 + \beta_1 X + \eta, Z] = \beta_1 \,\mathrm{Cov}[X, Z],$$

so

$$\beta_1 = \frac{\mathrm{Cov}[Y, Z]}{\mathrm{Cov}[X, Z]} = \frac{\mathrm{Cov}[Y, Z]/\,\mathrm{Var}[Z]}{\mathrm{Cov}[X, Z]/\,\mathrm{Var}[Z]} = \frac{\text{reduced form}}{\text{first stage}}.$$

**2SLS derivation.** Substituting $X$ into the outcome model,

$$\begin{aligned} Y &= \beta_1(\pi_{22} W + \pi_{21} Z + \pi_{20} + \varepsilon_2) + \beta_2 W + \beta_0 + \varepsilon \\ &= \beta_1 \pi_{21} Z + (\beta_2 + \beta_1 \pi_{22}) W + \underbrace{(\beta_1(\pi_{20} + \varepsilon_2) + \beta_0 + \varepsilon)}_{\xi}, \end{aligned}$$

hence

$$Y = \beta_1(\pi_{22} W + \pi_{21} Z) + \beta_2 W + \xi = \beta_1 \underbrace{\mathbb{E}[X \mid W, Z]}_{\text{population analogue of } \hat{X}} + \beta_2 W + \xi,$$

where $\pi_{20}$ is absorbed into $\xi$, so $\mathbb{E}[X \mid W, Z] - \pi_{20} = \pi_{21} Z + \pi_{22} W$ (intercept-free). Regressing $Y$ on $\mathbb{E}[X \mid W, Z]$ and $W$ recovers $\beta_1$; the sample analogue replaces it by the first-stage fitted value $\hat{X}$.

#### 7.2.2.4 From linear to nonparametric IV

The linear analysis assumes a constant effect $\beta_1$ for everyone, and then IV = ILS = 2SLS. If effects are heterogeneous ($Y_i(1) - Y_i(0)$ varies), what does the IV estimand identify? A key setting is the randomized experiment with non-compliance, in which random assignment $Z$ instruments treatment received $X$; Angrist, Imbens and Rubin (1996) give a potential-outcome framework.

### 7.2.3 Randomized experiments with non-compliance

**Motivation.** Identify the effect for compliers under random assignment, monotonicity and exclusion.

#### 7.2.3.1 Setting

A drug is tested in an RCT: $Z$ = random assignment, $X$ = treatment actually received, $Y$ = health outcome. Some patients assigned to treatment refuse the drug; some controls obtain it elsewhere. Comparing $X = 1$ with $X = 0$ is confounded (compliance is self-selected). Use $Z$ as an instrument: it is randomized (independent of potential outcomes) and affects $Y$ only through $X$ (exclusion).

**Potential outcomes** (Angrist, Imbens and Rubin 1996, JASA). Outcomes are indexed by assignment: $Y(z)$, $z = 0, 1$. Treatment received is a post-assignment variable with potential values $X(z)$. Observed data: $Z$, $X = X(Z)$, $Y = Y(Z)$. Units are divided into latent *compliance types* $S = (X(0), X(1))$:

| | $X(0) = 0$ | $X(0) = 1$ |
|---|---|---|
| $X(1) = 0$ | never-taker ($X = 0$ always) | defier ($X$ opposes $Z$) |
| $X(1) = 1$ | complier ($X$ follows $Z$) | always-taker ($X = 1$ always) |

Only compliers change treatment when $Z$ changes. $S$ is latent, and each observed $(Z, X)$ cell is a mixture: $(0,0)$: {C, NT}; $(0,1)$: {AT, D}; $(1,0)$: {NT, D}; $(1,1)$: {C, AT}. Additional assumptions are needed to identify type-specific effects. Goal: CACE $= \mathbb{E}[Y(1) - Y(0) \mid \text{complier}] = \text{ITT}_y / \text{ITT}_x$.

#### 7.2.3.2 Intent-to-treat effects

$S$ does not depend on $Z$; it is a baseline characteristic. Define type-specific ITT effects $\text{ITT}_s = \mathbb{E}[Y(1) - Y(0) \mid S = s]$, $s \in \{c, n, a, d\}$. With $\pi_s$ the proportion of type $s$, the global ITT $\text{ITT}_y := \mathbb{E}[Y(1) - Y(0)]$ is

$$\text{ITT}_y = \pi_c \text{ITT}_c + \pi_n \text{ITT}_n + \pi_a \text{ITT}_a + \pi_d \text{ITT}_d.$$

#### 7.2.3.3 Identification assumptions

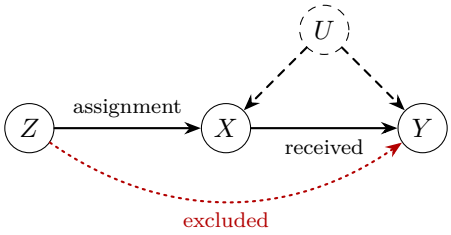



**Figure.** *Randomized encouragement with non-compliance. $Z$ is randomized (A1), affects $Y$ only through $X$ (exclusion, A2: the dotted arrow is absent), and no one does the opposite of the assignment (monotonicity, A3). Unobserved $U$ confounds treatment received and outcome.*

**A1 (random assignment)** $\{Y(0), Y(1), X(0), X(1)\} \perp\!\!\!\perp Z$.

**A2 (strong monotonicity)** (1) no defiers: $X(1) \geq X(0)$ for all units;

(2) compliers exist: $\mathbb{E}[X \mid Z = 1] \neq \mathbb{E}[X \mid Z = 0]$. Monotonicity often holds by design; with one-sided compliance $X(0) = 0$.

**A3 (exclusion restriction for non-compliers)** $Y(0) = Y(1)$ for all units with $S \in \{n, a\}$. For non-compliers $X$ is the same under both assignments, yet $Z$ could still affect $Y$ directly (e.g. a placebo effect); A3 rules out such direct $Z \to Y$ paths, the same "no direct effect" assumption as in linear IV, phrased in potential-outcome language.

#### 7.2.3.4 Identification

A2 gives $\pi_d = 0$; A3 gives $\text{ITT}_n = \text{ITT}_a = 0$. Hence

$$\text{ITT}_y = \pi_c \text{ITT}_c + \pi_n \cdot 0 + \pi_a \cdot 0 + 0 \cdot \text{ITT}_d = \pi_c \text{ITT}_c, \qquad \text{ITT}_c = \text{ITT}_y / \pi_c.$$

With $0 < \pi_c < 1$, $|\text{ITT}_y| < |\text{ITT}_c|$: the global ITT is a conservative (attenuated) estimate.

**Compliance-type proportions.** Under A1 and A2, when $Z = 0$ only always-takers have $X = 1$, and when $Z = 1$ only never-takers have $X = 0$:

$$\pi_a = P(X = 1 \mid Z = 0) = \mathbb{E}[X \mid Z = 0], \qquad \pi_n = P(X = 0 \mid Z = 1) = 1 - \mathbb{E}[X \mid Z = 1],$$

$$\pi_c = 1 - \pi_a - \pi_n = \mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0].$$

**Observed cells under monotonicity.** $(Z, X) = (0, 0)$: compliers (with $X(0) = 0$) and never-takers; $(0, 1)$: always-takers only; $(1, 0)$: never-takers only; $(1, 1)$: compliers and always-takers. Two cells are pure and directly identify non-complier means; two are mixed, but the complier means can be solved for using the known proportions:

$$\mathbb{E}[Y \mid X = 0, Z = 0] = \tfrac{\pi_c}{\pi_c + \pi_n} \mathbb{E}[Y(0) \mid \text{complier}] + \tfrac{\pi_n}{\pi_c + \pi_n} \mathbb{E}[Y(0) \mid \text{never-taker}],$$

$$\mathbb{E}[Y \mid X = 0, Z = 1] = \mathbb{E}[Y(0) \mid \text{never-taker}], \qquad \mathbb{E}[Y \mid X = 1, Z = 0] = \mathbb{E}[Y(1) \mid \text{always-taker}],$$

$$\mathbb{E}[Y \mid X = 1, Z = 1] = \tfrac{\pi_c}{\pi_c + \pi_a} \mathbb{E}[Y(1) \mid \text{complier}] + \tfrac{\pi_a}{\pi_c + \pi_a} \mathbb{E}[Y(1) \mid \text{always-taker}].$$

(Here $Y(0)$, $Y(1)$ denote the outcomes under the treatment actually received, which for compliers coincide with the assignment-indexed ones.)

**ITT on treatment.** $\text{ITT}_x = \mathbb{E}[X(1) - X(0)] = \pi_c$ by randomization: under monotonicity only compliers switch, so the fraction who switch is exactly $\pi_c$.

**Theorem 7.1 (CACE / LATE; Imbens and Angrist 1994, Angrist et al. 1996)**

Under A1–A3,

$$\text{CACE} \equiv \mathbb{E}[Y(1) - Y(0) \mid S = c] = \frac{\text{ITT}_y}{\text{ITT}_x} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0]}.$$

**Proof [Supplement].** By randomization (A1), $\text{ITT}_y = \mathbb{E}[Y(1, X(1)) - Y(0, X(0))]$. Split by compliance type $S \in \{c, a, n, d\}$. For always-takers and never-takers $X(1) = X(0)$, so by exclusion (A2) $Y(1, X(1)) = Y(0, X(0))$ and they contribute zero; by monotonicity (A3) there are no defiers. Hence $\text{ITT}_y = P(S = c)\, \mathbb{E}[Y(1) - Y(0) \mid S = c]$. Similarly $\text{ITT}_x = \mathbb{E}[X(1) - X(0)] = P(S = c)$, since $X(1) - X(0) = 1$ for compliers and 0 otherwise. Dividing (with $P(S = c) > 0$) gives $\text{CACE} = \text{ITT}_y / \text{ITT}_x$, and randomization identifies each ITT by the difference of observed means. ■

The CACE (local average treatment effect, LATE) is a ratio of two ITT effects; it equals the IV (Wald) estimand when random assignment is viewed as the instrument.

#### 7.2.3.5 A worked example

$N = 1{,}000$ subjects, 500 assigned $Z = 1$ and 500 $Z = 0$.

1. **Proportions.** In the $Z = 0$ group 100 take the drug: $\hat{\pi}_a = 100/500 = 0.20$. In the $Z = 1$ group 150 refuse: $\hat{\pi}_n = 150/500 = 0.30$. So $\hat{\pi}_c = 1 - 0.20 - 0.30 = 0.50$.
2. **ITT effects.** $\bar{Y}_{Z=1} = 7.0$, $\bar{Y}_{Z=0} = 6.0$, so $\text{ITT}_y = 1.0$; $\bar{X}_{Z=1} = 0.70$, $\bar{X}_{Z=0} = 0.20$, so $\text{ITT}_x = 0.50 = \hat{\pi}_c$.
3. **CACE** $= \text{ITT}_y/\text{ITT}_x = 1.0/0.50 = 2.0$.

The ITT (1.0) is attenuated by non-compliance, and the naive comparison of $X = 1$ with $X = 0$ would be confounded.

#### 7.2.3.6 CACE estimator

For two-sided non-compliance,

$$\widehat{\text{CACE}} = \frac{\widehat{\text{ITT}}_y}{\hat{\pi}_c} = \frac{\sum_i Y_i Z_i / \sum_i Z_i - \sum_i Y_i(1 - Z_i) / \sum_i (1 - Z_i)}{1 - \sum_i X_i(1 - Z_i) / \sum_i (1 - Z_i) - \sum_i (1 - X_i) Z_i / \sum_i Z_i},$$

with denominator $\widehat{\text{ITT}}_x = \hat{\pi}_c = 1 - \hat{\pi}_a - \hat{\pi}_n$. Standard errors via asymptotics or the bootstrap. Without A1–A3, nonparametric bounds can be obtained (Chapter 5) but are often too wide to be informative.

#### 7.2.3.7 The "fourth condition": homogeneity

Under homogeneity the IV estimand identifies the ATE for the whole population. Three progressively weaker versions, any one sufficient:

(a) a constant effect for all units, $Y(1) - Y(0) = \beta_1$;

(b) equal average effects across $Z$, $\mathbb{E}[Y(1) - Y(0) \mid Z = 0] = \mathbb{E}[Y(1) - Y(0) \mid Z = 1]$;

(c) no effect modification by $U$, $\mathbb{E}[Y(1) - Y(0) \mid U]$ constant.

(a) implies (b) implies (c).

(a) is rarely plausible; (b), (c) are weaker but still strong. Most applied work relies on monotonicity and the LATE interpretation, which needs only "no defiers" but identifies effects for compliers only (Hernán and Robins 2020, §§16.3–16.4).

#### 7.2.3.8 Summary

- **Linear IV:** $\beta_1 = \text{Cov}(Y, Z)/\text{Cov}(X, Z)$; IV = ILS = 2SLS under just-identification; assumes a constant effect.
- **Potential-outcome IV / LATE:** under A1–A3, CACE $= \text{ITT}_y/\text{ITT}_x$, the average effect for compliers; heterogeneous effects allowed, at the price of a local estimand.
- Under constant effects, CACE = ATE $= \beta_1$.

### 7.2.4 Appendix: Candidate instruments in observational studies

| Type | Example | Key concern |
|---|---|---|
| Genetic variant (Mendelian randomization) | ALDH2 polymorphism, alcohol metabolism | pleiotropy (direct effect on $Y$?) |
| Provider preference | physician's prescribing tendency | unmeasured patient factors drive referral |
| Geographic/temporal access | distance to facility, policy change date | correlated with SES, urban/rural |

Only relevance is empirically testable; exclusion and no confounding require subject-matter knowledge. Always ask whether $Z$ affects $Y$ through any path other than $X$ (*What If*, Fine Point 16.1).

### 7.2.5 Appendix: IV versus other causal methods

| IV estimation | IPW / g-methods |
|---|---|
| Does not require conditional exchangeability $Y(x) \perp\!\!\!\perp X \mid W$; works with unmeasured confounding. | Requires conditional exchangeability (all confounders measured). |
| Sensitive to violations of exclusion/no confounding, with bias amplification. | Minor violations give minor bias (no amplification). |
| Identifies only a local effect (compliers) under monotonicity. | Identifies the population ATE; extends to time-varying treatments. |

Prefer IV when unmeasured confounding is likely, a strong instrument exists, treatment is dichotomous, and either homogeneity or monotonicity is plausible (*What If*, §16.6).

### 7.2.6 Appendix: Practical issues with classic IV studies

**Quarter of birth** (Angrist and Krueger 1991): weak instrument (QOB explains $< 0.1\%$ of schooling variation); many-instruments bias (180 instruments push 2SLS toward OLS; Bound et al. 1995 obtained similar results with random data); exclusion (birth season may directly affect outcomes via health and school readiness); exogeneity (birth timing correlates with family background; Buckles and Hungerman 2013). Despite the critiques, influential in establishing the IV/LATE framework.
**Proximity to college** (Card 1995): exclusion (proximity correlates with urban/rural status, SES and school quality); monotonicity (living near a college may not uniformly increase schooling); LATE interpretation (the effect applies to compliers, those marginal with respect to proximity, likely lower-income, so LATE $\neq$ ATE) (Carneiro et al. 2011).

### 7.2.7 Appendix: Weak instruments

**Three problems.** (1) Wide confidence intervals: a weak $Z$–$X$ association gives large variance of $\hat{\beta}_{1,\text{iv}}$.

(2) Bias amplification: $\hat{\beta}_1 = \text{ITT}_y/\text{ITT}_x$, so a small bias in the numerator from a minor exclusion violation is divided by a small denominator and becomes large.

(3) Finite-sample bias: even with a valid instrument, 2SLS is biased toward OLS in finite samples. Rule of thumb: first-stage $F < 10$ (the $F$-test of $H_0 : \alpha = 0$ in $X = \alpha_0 + \alpha Z + \epsilon$) indicates a weak instrument (Staiger and Stock 1997; *What If*, §16.5).

**Example** (*What If*, §16.2): effect of smoking cessation $X$ on weight gain $Y$ with cigarette price in the state of birth as instrument $Z$. $P(X = 1 \mid Z = 1) - P(X = 1 \mid Z = 0) = 25.8\% - 19.5\% = 6.3\%$

(weak), first-stage $F = 0.8 \ll 10$; IV estimate 2.4 kg with 95% CI $[-36.5, 41.3]$: uninformative. A weak instrument with even slight exclusion violations can produce arbitrarily large bias.

**What to do.** Report the first-stage $F$ and compare it with the threshold for the number of instruments $K$ (Stock and Yogo 2005). Use confidence sets valid under weak instruments: the Anderson–Rubin (AR) set inverts a test that bypasses the first stage, does not rely on instrument strength, and with a single instrument is optimal, so there is no reason not to report it. With multiple instruments prefer LIML (limited-information maximum likelihood) over 2SLS: it is less biased with many or weak instruments (Anderson and Rubin 1949; Andrews, Stock and Sun 2019; Montiel Olea and Pflueger 2013).

**What not to do.** Screening on $F > 10$ and reporting only passing specifications (pretesting worsens size distortion); trying many instrument sets and picking the highest $F$ (instrument shopping, like $p$-hacking); bootstrapping IV inference (the bootstrap cannot handle the non-standard distribution under weak instruments); trusting conventional intervals $\hat{\beta} \pm 1.96\,\mathrm{SE}$ when $F$ is small (normality fails). Weak instruments are a research-design problem, not just an estimation problem; the best solution is a stronger instrument.

### 7.2.8 Appendix: Many instruments and robust IV

With $K$ instruments each gives $\hat{\beta}_k = \mathrm{Cov}(Y, Z_k)/\mathrm{Cov}(X, Z_k)$. If all are valid they agree on $\beta_1$; if some are invalid (e.g. pleiotropy) the estimates diverge, and 2SLS, which mixes them all, is contaminated. Robust alternatives: the weighted median (Bowden et al. 2016), consistent if more than 50% of instruments are valid; the mode-based estimator (Hartwig et al. 2017), consistent if a plurality agree; MR-Egger (Bowden et al. 2015), which allows directional pleiotropy through an intercept that corrects the bias. These are widely used in Mendelian randomization, where hundreds of genetic variants serve as instruments and some violate the exclusion restriction.